

CONCERNING LINES

3-1 INTRODUCTION

We have seen in Chapter 1 how from our everyday experience it is natural to imagine the existence of things called points and lines. Even though one cannot “see” a point or a line, the study of such imaginary objects turns out to be practically useful as well as interesting.

Because the objects of study in geometry are inventions of the mind, we must state very carefully the properties we wish them to possess. Then if we start with enough information about points and lines, other properties of points and lines can be *proved*. Many of these provable properties will be surprising, and not at all obvious at the beginning. It is, of course, impossible to prove anything about points and lines unless we agree in advance about some properties that they are to have. These agreed-upon properties, or assumed properties, are called *postulates*, or *axioms*.^{*} We begin the proper study of plane geometry in this chapter when we list our beginning postulates in Section 3-3. The postulates should be simple enough to seem almost obvious and yet must be sufficient for us to prove geometric facts that

seem to be true in our everyday life.

A different method of procedure would be to list all the known geometric facts and assume that they are true. Then when we had memorized all these facts, our study of geometry would be completed. There are at least three objections to such a plan. In the first place, we might not have listed all the facts needed and we would have no experience in finding new ones. In the second place, it would be a terrible strain upon the memory. To avoid memorizing everything, we need to study how different facts are related and how the more complicated ones depend upon other simpler ones. When we prove theorems from postulates or other theorems, we are studying the way the geometric facts are related to each other; this helps us to remember them. In the third place, we would lose the great amount of pleasure that we can get out of proving from very simple and obvious assumptions that certain other things are true. It is because of this pleasure that geometry has been so attractive to men since the time of Euclid. The fact that all geometry can be established from a small list of postulates gives to geometry a kind of beauty that has appealed to men as different as Abraham Lincoln and the Greek philosopher Plato.

3-2 STRAIGHTNESS

Soon, when we list our postulates, we shall be talking about *lines*, and of course what we have in mind are things which are like the familiar *straight* lines of our experience. But we will call them simply "*lines*." There is really nothing to be gained by putting in the extra word "*straight*" because it would add nothing to our use of the postulates. It will turn out much later that our lines *are* straight in the sense that any path between two points, other

* The word "postulate" is derived from a Latin word meaning "demand." "Axiom" comes from a Greek word meaning "worthy."

than a line, would be longer than the line.

When we draw pictures of lines we will, of course, draw them to *appear* straight, for this will help us see how the points and lines are related. But we will not be concerned with the straightness of the lines because the figures are meant to be only a guide to the proofs and to help us see what is going on. *The proofs are all completely independent of the figures.* By this we mean that we must not say something is true because of the appearance of a figure; rather, we must give reasons for its truth. The things that are true in our geometry are only those things which can be proved by our use of our assumptions and laws of logic. But as we progress in our study of geometry we will gradually use figures more and more freely when it is quite clear how to fill in a complete argument.

To understand why it is that we are not concerned with straightness at all, at least to begin with, we shall look at a plane in two different ways. First, imagine a sheet of paper with lines drawn on it so that some of the lines might look like Fig. 3-1. Now suppose you have a pane of uneven glass between your eyes and the paper, so that when you look at the drawing it looks like Fig. 3-2. Note that in looking at the drawing, with or without the glass, you are seeing the same lines. In one case they appear straight; in the other case they do not. In the first figure A and B may appear to be the same distance apart as A' and B' , but this may not be so in the second figure. Therefore, *if it were possible to see the plane only through the glass, you would have to be told which sets of points are "straight" lines, and whether or not A and B are the same distance apart as A' and B' .*

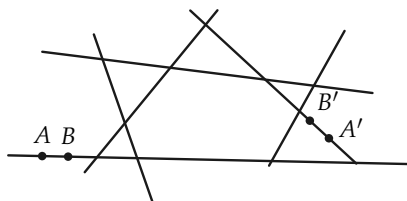


FIGURE 3-1

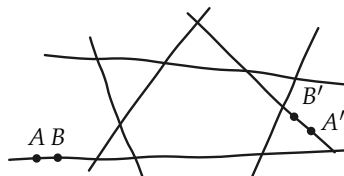


FIGURE 3-2

In our study of geometry we shall be talking about imaginary things—points and lines. We cannot really *see* our lines. Hence we shall not call them “straight” but refer to them only as *lines*. All the properties that we need to know about these lines will be stated clearly in the postulates.

3-3 THE POSTULATES

We imagine that there is a *set* (or “bunch,” or “collection”) of things which we call *points*. This set of points will be called *the plane*. In the plane certain other sets of points are called *lines*. We represent points by capital letters and lines by small letters. When a point P is one of the points of the line l (Fig. 3-3), we say that “ P is on l ” and also that the line “goes through P ,” or “passes through P ,” or “ l contains P .” If two lines l and m have a point P in common (Fig. 3-4), that is, if P is on both l and m , then we say that the lines *intersect*. Sometimes we will label two points with the same letter but different *subscripts*, such as P_1 and P_2 . We read P_1 as “ P sub-one” and P_2 as “ P sub-two.”

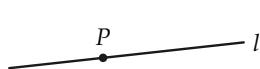


FIGURE 3-3

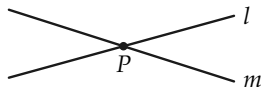


FIGURE 3-4

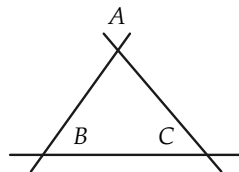


FIGURE 3-5

The postulates will be divided into six groups, of which only the first group will be given in this section. We shall give the other groups as we need them. It is extremely important at the beginning of your study of geometry that you understand the relationship between our mathematical postulates and the real world. We decide *what* we are going to assume about the imaginary points and lines of our geometry by *looking* at the real world. As a matter of fact, in order to choose the postulates for our geometry, we need only to look at some simple drawings. For example, the postulates in our first group are suggested by Fig. 3–5. Study this figure to see if you can decide what these postulates will be.

Of course, after we have decided what postulates to use, we must prove our theorems by *using these postulates and not by looking at other figures*.

Postulate Group I. This group contains three postulates, indicated by I–1, I–2, and I–3.

I–1 *The three-point postulate.* There are at least three points which do not all lie on one line.

I–2 *The point-line postulate.* For any two different points there is exactly one line containing these points.

I–3 *The line-point postulate.* Every line contains at least two points.

The point-line postulate tells us one way we want lines to behave. The line-point postulate *starts* us on the way to getting points on a line. The three-point postulate tells us that not all points are on one line. It gets us off the line.

With only these postulates we cannot as yet prove very much. The following two theorems are about all that can be formulated.

Theorem 3–1. *Two different lines do not intersect in more than one point.*

Proof. The theorem is either true or false. We shall prove that it cannot be false. Hence it must be true. We therefore suppose that the lines intersect in at least two points and then show that this is impossible. Suppose the lines are l and m and that they intersect in the two points A and B (Fig. 3-6). We would then have two different lines, l and m , through A and B . This contradicts the point-line postulate. Therefore our assumption that l and m intersect in more than one point is false and our theorem is true.

Remark. Several ideas from logic are involved in this proof. Comment on them.

Theorem 3-2. *There are at least three lines in the plane.*



FIGURE 3-6

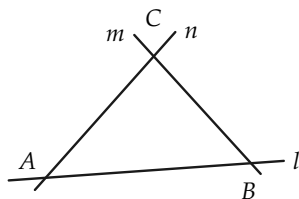


FIGURE 3-7

Proof.

STATEMENTS	REASONS
1. There are at least three points in the plane not all on one line. Call them A , B , and C .	Three-point postulate.
2. Let l be the line through A and B , m the line through B and C , and n the line through C and A .	Point-line postulate.
3. The three lines l , m , and n are all different from one another.	If any two were the same line, all three points A , B , and C would be on one line, contrary to the way they were chosen.

The three points and lines are shown in Fig. 3-7. Of course, it looks just like the drawing (Fig. 3-5) that suggested our first postulates.

Remark. At this point we cannot prove from our postulates that there are any more points or lines than shown in Fig. 3-7.

Exercise Group 3-1

1. Restate the point-line and the three-point postulates in other words.
2. Prove that if l is any line, there is at least one point not on l .
3. Explain why both a and b in Fig. 3-8 cannot represent lines if P and Q represent points.



FIGURE 3-8

4. Prove that if l is a line and P is on l then there is another point on l .
5. Why can we not prove there

- are more than three lines in our plane?
6. State what "intersect" means.
 7. Suppose that l is a line. List all the things you can say about this line and points of the plane. Give your reasons.
 8. Given Fig. 3-9, what can you conclude about l and m ? Why?
 9. A line l contains the points P and Q . The line m also con-
 10. Two lines have a point A in common. What can you say about the lines?
 11. Prove that there are at least two lines on each point.
 12. Why, when looking at Fig. 3-10, can we not prove that there are at least five points in the plane?

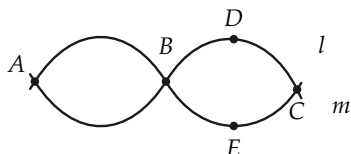


FIGURE 3-9

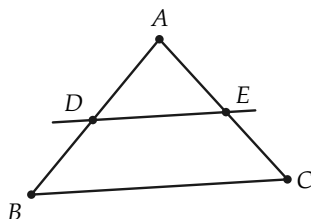


FIGURE 3-10

3-4 POSTULATES OF BETWEENNESS

We come now to our assumptions about the order, or arrangement, of points on a line. It is interesting that Euclid did not state these assumptions, but used them, without comment, wherever necessary. It was the German mathematician Moritz Pasch (pronounced "posh") who first, in the latter part of the 19th century, made a study of the properties of betweenness actually needed in geometry. The postulates given below are a modification of those of Pasch. You will have observed that the postulates of Group I seemed very obvious, and perhaps those of Group II will also. But remember that we are trying to state clearly what we imagine to be true about a plane, and hence we must be careful to state *everything* we are going to use.

Postulate Group II. Betweenness. There is a relation, called *between*, such that one of three points is sometimes between the other two. Of course, if we consider *any* three points, it does not *have* to happen that any one is between the others. But if for three points A , B , and C it is true that B is between A and C , we shall indicate this by writing " ABC " or " CBA ," that is, with B in the middle.

II-1 *The ABC betweenness postulate.* If ABC , that is, if B is between A and C , then A , B , and C are three different (distinct) points on a line. (A drawing of the line and points might look like Fig. 3-11.)



FIGURE 3-11

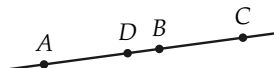


FIGURE 3-12

II-2 *The three-point betweenness postulate.* For every three points on a line, exactly one of them is between the other two. That is, if R , S , and T are on one line, then exactly one of the statements RST , STR , and TRS is true.

II-3 *The four-point betweenness postulate.* Any four points on a line may be labeled A_1 , A_2 , A_3 , A_4 so that the only betweenness relations are the same as the order of the subscript numbers. That is, the only betweenness relations are $A_1A_2A_3$, $A_1A_2A_4$, $A_1A_3A_4$, $A_2A_3A_4$. Of course these may all be reversed as $A_3A_2A_1$, $A_4A_2A_1$, $A_4A_3A_1$, $A_4A_3A_2$. A similar statement is to hold for any five points on a line, and in fact for any finite number of points on a line.

II-4 *The line-building postulate.* If A and B are two points, then there is at least one point C such that ABC , and at least one point D such that ADB . (See Fig. 3-12.)

Before stating the last of the betweenness postulates, we need some definitions.

Definition 3-1. Two points not on a line l are said to be *on the same side of l* if no point of l is between the two points. (See Fig. 3-13.) Two points not on a line l are said to be *on opposite sides of l* if there is a point of l between the two points. (See Fig. 3-14.)

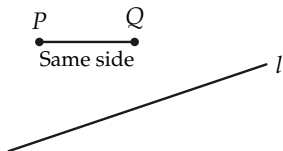


FIGURE 3-13

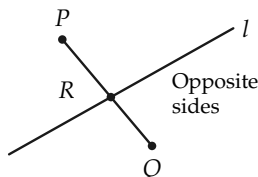


FIGURE 3-14

II-5 *The two-side postulate.* Each line separates the plane. By this we mean that all points of the plane not on the line are divided into two sets, called the two sides of the line, such that if P and Q belong to one of these sets, then P and Q are on the same side of the line, whereas if P belongs to one set and Q belongs to the other, then P and Q are on opposite sides of the line. (Briefly, this says that a line has two sides.)

Before using these postulates, let us first discuss their meaning in everyday life. If three men are standing in a line (Fig. 3-15) and A cannot see C because B is in the way, then we say that B is between A and C . On the other hand, B and C can see each other perfectly. Hence A is not between B and C . That is, we do not have BAC .

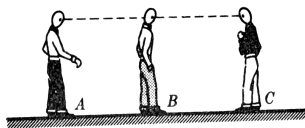


FIGURE 3-15

The ABC betweenness postulate says that to talk about betweenness, we must have the men “lined up.” The three-point betweenness postulate says that when three men are lined up, exactly one of them is between the other two.

The four-point betweenness postulate gives a definite meaning to what we expect to see when we say, “Line up these four people.” It tells exactly who is between whom. An easy way to remember the content of this postulate is to observe that of four points on a line, two of them are never between any others. These are called the *endpoints*. The other two points are between points exactly twice.

The line-building postulate tells us first that there is always a point C “beyond” B from A . In fact, Euclid stated (vaguely) something very much like this as a postulate. But our postulate also tells us that there are points between any two. Euclid never stated this. It is this postulate that gives the existence of more than two points on a line, and therefore it is called the line-building postulate. You might think about the relation between this postulate and the real world. When you draw pictures of points, can you always mark a picture of a third point between any two?

The two-side postulate says that a line has two sides. It merely expresses the way we tell whether or not two of us are on the same or different sides of a fence. If, when we look at each other, we see the fence between us, we know we are on opposite sides of it. If we do not see the fence between us, we are on the same side.

Now, using these postulates, we can define some familiar geometric objects.

Definition 3–2. If A and B are any two points, then the set of points consisting of A and B and all the points between A and B will be called the *line segment* (or *segment*) determined by A and B . The segment will be indicated by “ AB .” The points A and B

are called the *ends* of the segment. In Fig. 3-16, P is a point of the segment AB .

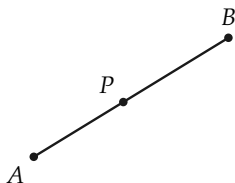


FIGURE 3-16

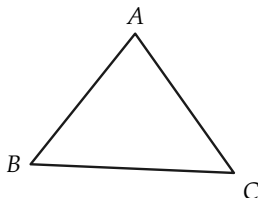


FIGURE 3-17

Definition 3-3. If A , B , and C are three points not all on one line,* the set of points of the segments AB , BC , AC is called a *triangle* (Fig. 3-17). We call the points A , B , and C *vertices* of the triangle and the segments AB , AC , BC *sides* of the triangle. To indicate “triangle ABC ,” we write “ $\triangle ABC$.”

Before working the exercises, read all the postulates again to be sure that you understand them.

Exercise Group 3-2

1. Draw a line and mark five points on it. Write out all the betweenness relations.
2. If ABC and ACD , prove that the four points A , B , C , and D lie on one line.
3. Give in your own words the definitions of “segment,” “triangle,” and “two points are on the same side of a line.”
4. If A , B , C , and D are four points of a line and we know that CAD and ADB , what are the other betweenness relations? Draw a picture. Is the picture proof?

* Points all lying on a line are said to be *collinear*. Hence in the definition we could have said, “If A , B , and C are not collinear ...”.

- *5. If A , B , C , and D are four points of a line and we know that CAD and CAB , what are the other betweenness relations?
6. If E , F , G , and H are four points on a line and we know that EFG , how many possible positions are there for H that can be described in terms of our betweenness relations?
7. A and B are points on a line l . Write three true statements about points either on l or not on l . Give reasons, as below.

STATEMENTS	REASONS
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- | | |
|---|------------------------------|
| 1. There is a point C such that ACB . | The line-building postulate. |
| 2. _____ | _____ |
| 3. _____ | _____ |
| 4. _____ | _____ |
8. If RST and VRS , relabel the points R , S , T , and V as A_1 , A_2 , A_3 , and A_4 (in some order) in accordance with II-3. If CAD , BDE , and BCA , relabel the points A_1 , A_2 , A_3 , A_4 , and A_5 in accordance with II-3.
9. Prove that there are at least four lines on each point.
10. If DEF and EGF , then the four-point betweenness postulate implies that _____ and _____.
11. If RST , then what postulate tells us that R , S , and T are distinct points on a line?
- *12. Draw a line and choose two points on it, A and B (Postulate I-3). Mark a point C such that ABC (Postulate II-4). Mark a point D such that BCD (Postulate II-4). Is it true that ACD ? Why? Use II-3 to prove that ACD .
- *13. Draw a line and choose two points on it, A and B . Choose a point C such that ACB (Postulate II-4). Now choose a point D such that CDB . Is it true that ACD ? that ADB ? Use II-3 to prove this. Now choose a point E such that DEB . What are all the betweenness relations between these five points? Can this process be indefinitely repeated? Describe, in your own words, the meaning of "this process." How many points are there on a line segment?
- *14. Draw a line and choose a point A on one side and a point B on the other side. Now mark a point C on the opposite side of the line from A . Points B and C will be on the same side of the line. How does this follow from II-5?
- *15. A line l (Fig. 3-18) does not contain any of the vertices of a triangle. The line l does inter-

sect the side AB , but does not intersect the side BC . Prove that l must intersect AC .

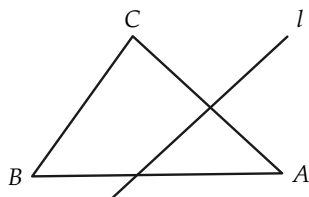


FIGURE 3-18

*16. Draw a line and choose two points A and B on one side of the line. Choose C so that ACB . Can you prove that C and A are on the same side of the line?

*17. If A is on line l and B is not on l and ACB , prove that C and B are on the same side of l .

*18. Prove that for every line l there is a point on each side of l .

3-5 RAYS AND ANGLES

It is a familiar fact from experience that a point O on a line l divides the other points of the line into two sets—those on one side and those on the other side of the point O on the line l . (See Fig. 3-19.) From our postulates on betweenness we can now make precise definitions of this observation. First we need to know that a point on a line separates the line.

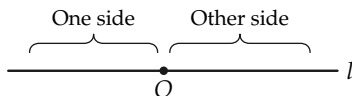


FIGURE 3-19

Theorem 3-3. The two-side theorem. *Let O be any point of a line l . The points of l , different from O , are separated into two sets such that: (1) If P and Q are in the same set, then either OPQ or OQP . (2) If P is in one set and Q is in the other set, then POQ .*

**Proof.* Choose a point A not on l and consider the line m through O and A (see Fig. 3-20). We are using I-1 and I-2. By the two-side

postulate, II-5, the line m separates the plane into the two sides of m . Let one of the sets, which we seek, consist of the points of l which are on one side of m , and let the other set consist of the points of l on the other side of m .

If P and Q are in one of these sets, they are by definition on the same side of m , and therefore O is not between them. Hence by II-2 we must have either OPQ or OQP . This proves (1).

Now suppose that P is in one set and Q in the other set (see Fig. 3-21). Then by definition of the sets, P and Q are on different sides of m . By the two-side postulate, II-5, there is a point of m between P and Q . But l and m have only point O in common. Hence the point between must be O , and we have POQ . This proves (2).

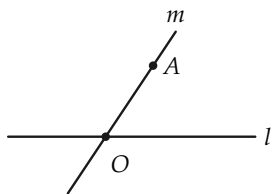


FIGURE 3-20

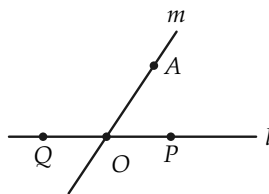


FIGURE 3-21

Remark. The two sets of the theorem are not dependent on how we choose the line m , because we end up with the same betweenness relations.

Definition 3-4. The two sets of Theorem 3-3 are called the two *sides* of O on the line l .

Definition 3-5. A *ray* from O , or a ray with end O , is a set of points consisting of O and all the points on one side of O of a line l through O (see Fig. 3-22).

* Starred proofs are difficult and may be omitted. However, the meaning of the theorems must be understood.

If P is any point of the ray, we may speak of “ray OP .” Note that we may describe the ray as the set of points consisting of O and P and all points Q such that OQP and all points R such that OPR . Refer back to Theorem 3-3.

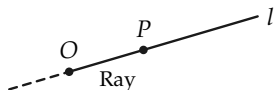


FIGURE 3-22

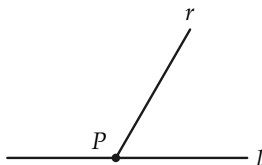


FIGURE 3-23

Exercise 3-3

Consider a line l and a ray r , not on l , from a point P on l , as in Fig. 3-23. What do we mean by “ r ’s side of l ”?

Definition 3-6. The set of all points on two rays from a point is called an *angle*. The point is called the *vertex* of the angle, and the two rays are called the *sides* of the angle. If the two rays are on the same line, the angle is called a *straight angle*. If O is the vertex and A and B are points other than O on the sides of the angle, we shall speak of “angle AOB ” and write this as “ $\angle AOB$.” (See Fig. 3-24.)

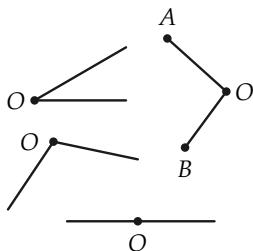


FIGURE 3-24

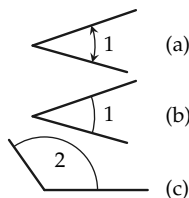


FIGURE 3-25

Remark. Note that what we call the angle is the set of points in the *sides* of the angle. Points “between” the rays are *not* part of the angle. Later we discuss the inside and the outside of an angle. It will be convenient to have a way to clearly indicate the sides of the angles. This can be done as in Fig. 3-25(a), and the angle can be numbered. Since the arrowheads indicating the sides are not vital, we can as well use the simpler notation shown in Fig. 3-25(b) and (c). Remember that this notation is just a way of indicating the sides of the angle.

Definition 3-7. If two lines l and m intersect at O and the four rays from O are as indicated in Fig. 3-26, then angles 1 and 2 are said to be *vertical* with respect to each other. For brevity, we say that 1 and 2 are a pair of vertical angles. Angles 3 and 4 are also vertical to each other.

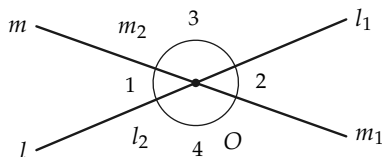


FIGURE 3-26

Definition 3-8. If two angles have a common vertex and a common side and if the other sides are on opposite sides of the line on the common side, then the angles are said to be *adjacent* to each other (Fig. 3-27).

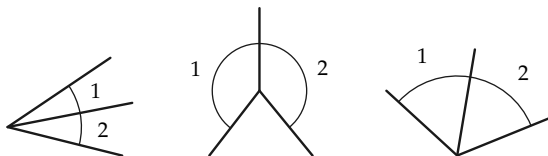


FIGURE 3-27

Definition 3-9. Two adjacent angles whose noncommon sides form a straight angle are said to be *supplementary and adjacent* to each other (Fig. 3-28).

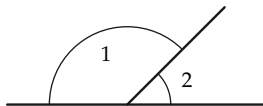


FIGURE 3-28

Exercise Group 3-4

1. State in your own words the definitions of "ray," of "angle," and of "straight angle."
2. How "long" are the sides of an angle?
3. If $\angle AOB$ and $\angle DOC$ are vertical to each other with B, O, D collinear, what kind of angle is $\angle AOC$?
4. If $\angle AOC$ is a straight angle and OR is a ray from O , how are angles AOR and ROC related?
5. Two lines intersect at O . With the four rays from O , how many different angles with vertex O can be found? Pick out the pairs of vertical angles.
6. Three lines intersect at O . With the six rays from O , how many different angles with vertex O can be formed? How many pairs of vertical angles are there?
7. Draw two angles with a common vertex but which are not adjacent to each other.
8. Draw two angles with a common vertex and side but which are not adjacent.
9. State Definitions 3-7, 3-8, and 3-9 in your own words.
10. If $\angle AOB$ is not a straight angle, how many angles are there that are vertical with respect to $\angle AOB$?
11. Why is it not sensible to say, " $\angle AOB$ is a vertical angle"?

3-6 INTERIOR AND EXTERIOR

When we draw a picture of an angle (not a straight angle) on a sheet of paper, we can tell by looking at it which points of the plane are inside and which are outside the *picture* of the angle. But in our development of geometry starting from but a few postulates, we must *define* what these sets are. (Why?) Definition 3-10 is one that can be understood by a blind person provided he knows what an angle is and that a line separates the plane. To appreciate the definition, consider Fig. 3-29. The point D lies on one side of l , which side has been shaded with vertical lines. The point C lies on one side of m , which side has been shaded with horizontal lines. The points in the plane which are shaded twice (i.e., where the vertical and horizontal lines cross) fill up the *interior* of $\angle COD$.

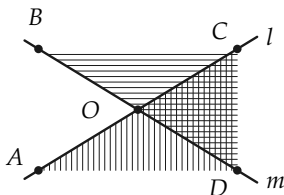


FIGURE 3-29

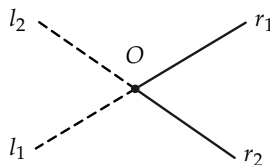


FIGURE 3-30

Exercise Group 3-5

- Copy the lines l and m and the points A, B, C, D on a sheet of paper and shade all the points of the plane which are (a) on the same side of l as B and the same side of m as A , (b) on the same side of l as B and on the same side of m as C , (c) on the same side of l as D and on the same side of m as A .
- In each case name the angle whose interior has been described.

Definition 3–10. Consider an angle, not a straight angle, with vertex O and whose sides are the rays r_1 and r_2 on the lines l_1 and l_2 , respectively. The *interior* of the angle is the set of all points on the same side of l_1 as r_2 and on the same side of l_2 as r_1 . The *exterior* of the angle is the set of all points in the plane which are not points of the angle and are not in the interior of the angle. (Fig. 3–30.)

There are several properties of the interior and exterior of angles that we can now prove. These properties are all quite obvious when we look at a figure. But even when something seems obvious from a picture, we must be sure that we have the right postulates so that our geometry agrees with our feelings about what should be true—that is, with the picture. It is really not at all obvious that our *geometry* agrees with our *picture*. Let us see that this is so by proving some “obvious” theorems.

In all these theorems we shall be concerned with an angle whose vertex is O , which is not a straight angle, and whose sides are the rays r_1 and r_2 .

***Theorem 3–4.** *If P and Q are both in the interior of $\angle O$, then the segment PQ is in the interior of $\angle O$.*

Proof. We must show that if A is any point between P and Q , then A is in the interior of the angle (Fig. 3–31). This is easy if we use the result of Exercise 16 in Exercise Group 3–2. By that exercise, since P and Q are on one side of l_1 , then A is also on that same side of l_1 . Similarly, A is on the same side of l_2 , as are both P and Q . Hence, by the definition of *interior* of an angle, A is in the interior of the angle formed by rays r_1 and r_2 .

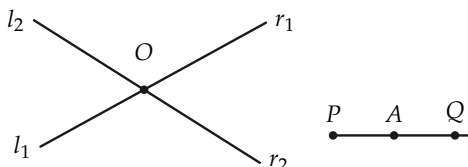


FIGURE 3-31

***Theorem 3-5.** *If r is a ray from O other than r_1 and r_2 , then either all the points of r other than O lie in the interior of $\angle O$ or they lie in the exterior of $\angle O$.*

Proof. Let r'_1 and r'_2 be the rays from O on l_1 and l_2 other than r_1 and r_2 . (See Fig. 3-32.) If $r = r'_1$ or $r = r'_2$, then r is in the exterior. If r is neither r'_1 nor r'_2 , then r lies on a definite side of l_1 and a definite side of l_2 because of the way a ray is defined. Hence r is in either the interior or exterior of $\angle O$.

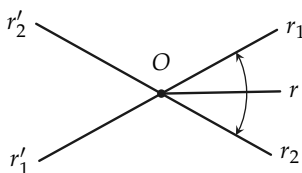


FIGURE 3-32

***Theorem 3-6.** *If P is in the interior of $\angle O$ and Q is in the exterior of $\angle O$, then the segment PQ contains a point of the angle.*

Proof. There are five possible locations of the point Q , represented by the points $Q_1, \dots, Q_5$ in Fig. 3-33. (Describe these positions in terms of sides of lines.) For positions Q_1, Q_2, Q_4, Q_5 the proofs of the theorem are quite easy and are assigned as exercises below. Hence we restrict our attention here to the position Q_3 .

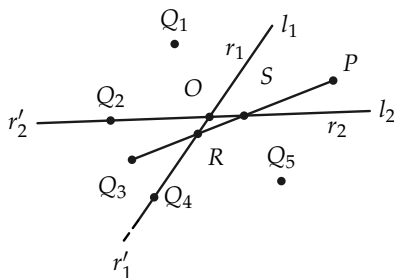


FIGURE 3-33

Since Q_3 is on opposite sides of both l_1 and l_2 from P (why?), the segment Q_3P must contain a point R of l_1 and a point S of l_2 . We assert that either R or S is a point of the angle, that is, a point of either r_1 or r_2 . If R is on r_1 , our segment contains a point of the angle. Suppose that R is not on r_1 . Then R is on r'_1 and hence has the same relative position as Q_4 . (Why?) But then the segment RP contains a point of the angle which must be S . (Why?) Since Q_3RP and RSP , we have Q_3SP . (Why?)

Definition 3-11. The *interior* (or *inside*) of a triangle is the set of all points which are interior to each of the angles of the triangle. The *exterior* of a triangle is the set of points which are neither on the triangle nor in the interior.

If the triangle is ABC (see Fig. 3-34), an alternative definition of interior would be "the set of all points on the same side of AB as C , on the same side of BC as A , and on the same side of CA as B ." Show that these two definitions say the same thing.

We can prove the following remark by using Theorem 3-6. The remark is an "obvious" theorem that is hard to prove. We will not need this result later.

Remark. If P is in the interior of $\triangle ABC$ and Q is in the exterior, then the segment PQ contains a point of the triangle (Fig. 3-35).

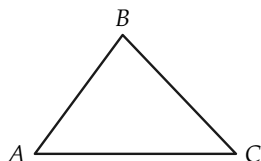


FIGURE 3-34

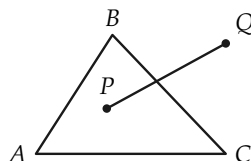


FIGURE 3-35

Exercise Group 3-6

- Referring to Fig. 3-36, draw an angle with vertex O . Pick a point A on one side of $\angle O$. Draw an angle with vertex A with one side the ray AO and the other side a ray intersecting the other side of $\angle O$ in B . Shade the interior of $\angle O$ and the interior of $\angle A$. Are points in both interiors also in the interior of $\triangle OBA$?

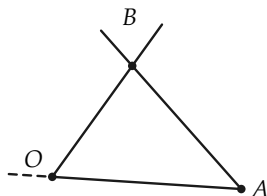


FIGURE 3-36

- Draw a $\triangle ABC$. If a point P is on the same side of AB as C and on the same side of BC as A , show that P is not necessarily on the same side of AC as B .
- Draw two angles whose interiors have no points in common.
- A straight angle does not have an interior. Tell why we do not try to define interior for a straight angle.
- State in your own words the definitions of "interior of an angle" and "interior of a triangle."
- For a $\triangle ABC$, the point P is in the interior of $\angle A$ and not in the interior of $\angle B$. Can you be sure that P is in the exterior of $\triangle ABC$?
- Consider the angle of Fig. 3-37, with sides l_1 and m_1 indicated on lines l and m . Tell where the point P is if:
 - P is on the opposite side of m from l_1 .
 - P is on the opposite side of l from m_1 .

- (c) P is on l_1 .
 (d) P is on m_1 .
 (e) P is on the same side of l as m_1 .
 (f) P is on the same side of l as m_1 and on the same side of m as l_1 .

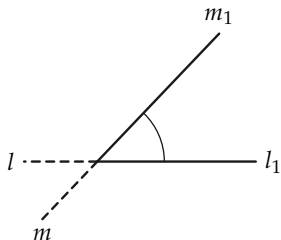


FIGURE 3-37

8. Prove that the following definition is equivalent to Definition 3-11: The interior of $\triangle ABC$ is the set of all points in the interior of $\angle A$ and also in the interior of $\angle B$. (With this definition it is not too difficult to prove the truth of the remark preceding this group of exercises.)
9. In $\angle AOB$, not a straight angle, point P is between A and B . Prove that P is in the interior of the angle.
10. In triangle ABC we have the relations AXB , AYC , and XZY . Prove that Z is in the interior of the triangle.
11. If $\angle AOB$ is not a straight angle and we have the relations

XOB and XYA , prove that Y is in the exterior of $\angle AOB$.

*12. In Theorem 3-6 suppose Q is in position Q_1 , that is, on the same side of l_2 as r_1 and on the opposite side of l_1 as r_2 (see Fig. 3-38). Show that there is a point R of r_1 between Q and P . [Hint: Certainly there is a point R of l_1 between Q and P . Show that it is impossible that R be on r'_1 .]

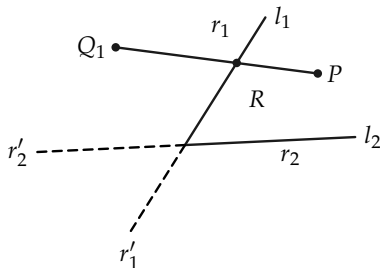


FIGURE 3-38

- *13. In Theorem 3-6 suppose Q is in position Q_2 , that is, on r'_2 (see Fig. 3-39). Show that there is a point R of r_1 between Q and P .

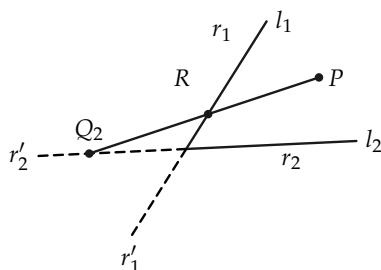


FIGURE 3-39

- *14. Tell why the positions Q_4 and Q_5 of Fig. 3-33 are much the same as positions Q_2 and Q_1 , respectively. If you have done positions Q_1 and Q_2 , why are positions Q_5 and Q_4 easy?

REVIEW OF CHAPTER 3

1. The fundamental relations of betweenness are expressed in the postulates. Study them again to be sure you know them.
2. Definitions of the following concepts were given in this chapter. Be sure that you know all of them.

two sides of a line (see II-5)	straight angle
segment	vertical angles
triangle	adjacent angles
collinear (see footnote to Def. 3-3)	supplementary angles
two sides of a point on a line	interior of an angle
ray	exterior of an angle
angle	interior of a triangle
	exterior of a triangle

3. The following statements have been proved in this chapter. First re-read them, and then study their proofs again.
 - (a) Two different lines intersect in at most one point.
 - (b) There are at least three lines.

- (c) A point O on a line separates the line into two sides.
- (d) If P and Q are in the interior of $\angle O$, then the segment PQ is in the interior of $\angle O$.
- (e) A ray from the vertex of an angle lies, except for the vertex, either in the interior or in the exterior of the angle.
- (f) If P is in the interior of $\angle O$ and Q is in the exterior, then the segment PQ intersects the angle.

Important Remark

By assuming a few properties of points and lines, we have been able to *define* many familiar types of figures and to *prove* that these figures have properties that may seem obvious. *Therefore we have proved that some of our figures agree with our geometry.* From now on in the book, we will often determine properties of betweenness by just looking at figures. The definitions and theorems of this chapter will enable us to use our figures in the confident belief that we could always go back and supply proofs. Therefore, we have completed the first stage in our development of geometry. The theorems proved in this chapter need not be memorized because the facts are all familiar from drawings. However, the postulates and definitions of this chapter are very important, and therefore should be memorized.

Review Questions

1. If A and B are distinct points, what can you say about a point P which is on the segment AB ? Why?
2. What postulate did we use to define the sides of a point on a line?
3. A line l contains the distinct points R and S . A line m also contains these points. What do you conclude?

4. If ABC , and B is on line l but A is not on l , is the point C on l ? Why?
5. Tell whether the following statements are true or false, and give explanations for your answers.
 - (a) For $\triangle ABC$, if the point S is in the interior of $\angle A$, $\angle B$, and $\angle C$, then it is a theorem that S is in the interior of the triangle.
 - (b) If APB , then P belongs to the segment AB , by definition.
 - (c) If ARQ and AGR , then AGQ .
 - (d) A point P not on an angle must be in either the interior or exterior.
 - (e) We postulate that a point on a line separates the line.
 - (f) If AQB , then A and B are on a ray from Q .
 - (g) A line segment contains a definite number of points.
6. What does it mean to say that three or more points are non-collinear? How can we be sure that there are such sets of points?
7. If ADB and BAC , what are the other betweenness relations?
8. Lines s and t intersect at Q forming rays s_1, s_2 and t_1, t_2 . Draw a figure and indicate the pairs of vertical angles. Which pairs of angles are adjacent? supplementary?
9. If A and B are on opposite sides of a line l , and B and C are on opposite sides of l , then A and C are on the same side of l . Why is this true?
10. Explain in your own words what it means for a point to be in the interior of an angle.

Review Test

1. Define the following: (a) angle, (b) segment, (c) when two points are on the same side of a line, (d) triangle.
2. Tell whether true or false:
 - (a) A, B, C are three points on a line and ABC and CAB .
 - (b) If PRU , then P, R , and U are distinct points on a line.
 - (c) It is a theorem that a line separates the plane.
 - (d) It is a theorem that a point separates a line.
 - (e) It is a theorem that there is a point between any two points.
 - (f) Betweenness is an undefined relation.
 - (g) If ABC , then CBA .
 - (h) If E, F, G are not on l and such that the segment EF does not intersect l , while the segment FG does intersect l , then the segment EG intersects l .
3. State in your own words:
 - (a) The point-line postulate.
 - (b) The ABC betweenness postulate.
 - (c) The line-building postulate.

ALGEBRA REVIEW

In this review you are supposed to use the properties of addition and multiplication of numbers (both positive and negative), along with the definitions of subtraction and division, to make simple proofs. All letters here refer to rational numbers (fractions). These exercises show how definitions are used in algebra.

Definition. If $a = b + x$, then x is called " a minus b " and is written " $x = a - b$." It is also called " b subtracted from a ."

1. If $c = p - q$, what is p ? Why?

2. Why does $7 - 3 = 4$? Write $7 - 3 = x$, and ask what property x must have.
3. Using the definition, what can you say about these numbers:
 $\frac{9}{5} - \frac{3}{5}$? $\frac{1}{2} - \frac{1}{3}$? $\frac{7}{8} - \frac{3}{4}$? $4.678 - 2.589$?

Definition. $-b$ (called “*negative b*”) is defined to be $0 - b$. That is, $-b = 0 - b$ and is a number such that $b + (-b) = 0$.

4. Why does $-7 = 0 - 7$? $-3 = 0 - 3$? $-\frac{1}{2} = 0 - \frac{1}{2}$?
5. Show that $9 + (-3) = 9 - 3$, $11 + (-8) = 11 - 8$, $\frac{5}{6} + (-\frac{1}{3}) = \frac{5}{6} - \frac{1}{3}$.
- *6. Prove that $a + (-b) = a - b$. [Hint: Let $a - b = x$ so that $a = b + x$. Then $a + (-b) = b + x + (-b)$.]
7. Show that $-(7 + 3) = (-7) + (-3)$, $-(2 + 7) = (-2) + (-7)$.
8. Show that $-(a + b) = (-a) + (-b)$.

Definition. If $a = bx$ and $b \neq 0$, then x is called “*a divided by b*” and is written “ $x = a \div b$ ” or “ $x = a/b$.”

9. Explain why in this last definition we required that $b \neq 0$.
 [Hint: Would x be definite otherwise?]
10. According to the definition, $\frac{1}{2}$ is a number such that $1 = 2(\frac{1}{2})$.
 What then are the numbers $\frac{3}{4}$, $-\frac{2}{3}$, $4/-5$, $\frac{7}{16}$, $\frac{5}{10}$, $\frac{1}{1}$, $\frac{2}{2}$?
11. Why does $b \cdot a/b = a$?
12. Prove that $a/b + c/b = (a + c)/b$.
13. Show that $\frac{1}{2} = \frac{2}{4}$. [Hint: Let $\frac{1}{2} = x$. Then $1 = 2 \cdot x$. Why? Hence $2 = 4x$. Why? Hence $\frac{2}{4} = x$. Why?]
14. Prove that if $b \neq 0$ and $c \neq 0$, then $a/b = ac/bc$.